# Professional Issues in IT

COURSE INSTRUCTOR: SANIA UROOJ

EMAIL ADDRESS: sania.urooj@nu.edu.pk

# Continuing Profession

**Profession vs Occupation**

- Commonly people use these two words interchangeably. Are they the words meaning the same? Are they synonyms?
- What is the meaning of occupation?
  - A person involved in a job and earning money is having that job as an occupation

    Example: A doctor teaching in a school, college or university has an occupation of teaching. A carpenter, a driver etc..

# Differences between Occupation and Profession

- A profession needs extensive training and specialized knowledge. On the other hand, an occupation does not need any extensive training.
- A profession can be called an occupation when a person is paid for his particular skills, and his deep knowledge. Persons engaged in an occupation are not paid for their knowledge, but only for what they produce.
- Unlike a person engaged in an occupation, a professional has to undergo higher education.
- A profession tends to be autonomous, whereas, for an occupation, no one has autonomous power; he or she is supervised by another person.
- Unlike occupation, a profession demands that the responsibility lies with the individual.
- A profession is guided through certain ethical codes, and regulated by certain statute.

# Characteristics of Profession

Getting to the definition of profession ; there is none called hard and fast definition. Thus professions are often associated with some characteristics briefed as following:

- **Mastery of an Esoteric Body of knowledge**: usually acquired through higher education
  - A member of the profession needs this body of knowledge in order to practice
  - A related characteristic of professions is that they often embrace a division between researchers and practitioners

# Characteristics……..

- **Autonomy**: Professionals generally have a great deal of autonomy, justified by having an esoteric knowledge
  - Professions typically have autonomy both at the collective level as well as in individual practice
  - Profession sets its own standards rather than taking orders or suggestion from outsiders
- **Formal Organization**: Generally there exists a professional organization that:
  - Controls admission to the profession and sets standards
  - Involved in licensing & expelling its individual members

# Characteristics……..

- **Code of ethics**
  - There is a code of ethics & members of professions must adhere to the code no matter what their employment context (take an oath)

And an important characteristic mostly taken for granted:-

- **Social function:** Professions generally fulfill an important social function and for the welfare of the society
  - Medicine  (promoting health)
  - lawyers not alone but play an important role in justice

# Applying the Characteristics of a Profession

**Imagine a professional engineer named Ali who works in civil engineering.**

**1. Mastery of Knowledge:** Ali has a degree in civil engineering and has passed the professional engineering exam. He has in-depth knowledge of structural design, materials science, and construction practices.

**2. Autonomy:** Ali works on designing bridges and is responsible for ensuring they are safe and functional. He has the autonomy to make critical decisions about the design without needing constant supervision.

**3. Formal Organization:** Ali is a member of the Institution of Civil Engineers (ICE), a professional body that sets the standards for civil engineering practice.

# Applying the Characteristics of a Profession

**Imagine a professional engineer named Ali who works in civil engineering.**

**4. Code of Ethics:** Ali follows the ICE's code of ethics, which requires him to prioritize public safety, avoid conflicts of interest, and maintain integrity in all his work.

**5. Social Function:** By designing safe and reliable bridges, Ali contributes to the welfare of society, ensuring that the infrastructure is safe for public use

# PROFESSIONAL BODIES

- A profession is typically organized into one or more professional bodies.
- A **professional association** (also called a **professional body**, **professional organization**, or **professional society**) is usually a nonprofit organization seeking to further a particular profession, the interests of individuals engaged in that profession and the public interest.
- A professional body is a group of people in a learned occupation who are entrusted with maintaining control or oversight of the practice of the occupation
- A professional body usually starts by a group of people coming together because of a shared interest in a particular type of activity.

# Functions of Professional bodies

- As the professional body matures, it is likely to develop a range of functions, of which the following are the most important:
- establishing a code of conduct to regulate the way members of the body behave in their professional lives and a disciplinary procedure to discipline members who breach(fail to follow or conduct) this code;
- establishing mechanisms for disseminating(spreading) knowledge of good practice and new developments to its members, typically through publications and conferences but increasingly also through the use of the worldwide web;
- setting standards of education and experience that must be met by people wishing to become members of the body;
- advising government and regulatory bodies about matters within its area of expertise.

# RESERVATION OF TITLE AND FUNCTION

The use of the name of the profession may be restricted to those people who are appropriately qualified. A restriction of this sort is called ***reservation of title***.

The law may state that certain activities are restricted to people with appropriate qualifications. This is called ***reservation of function***.

# Is Computing A Profession

**Where does computing fit in the process of professionalization?**

What does the field of computing look like when described in terms of the mentioned characteristics?

- The field of computing is so diverse and complex
- The range of those who are called computer professionals is extremely broad
- Nothing in common except that, in one way or another, they involve the use of computer
- At first glance, it seem that computing possesses all of these characteristics, though in complex ways.
- *Scenario: Consider two professionals—one a software developer at a tech startup, and the other an IT consultant for a large corporation. Both work with computers but in very different capacities, showcasing the diversity within the computing profession. The startup developer might have more autonomy in creative solutions, while the consultant follows strict guidelines to meet client needs.*

# Is Computing A Profession

- Computer professionals have mastered an esoteric body of knowledge
- Computer professionals have varying degrees of autonomy depending on where they work
- There is no single organization governing the profession of computing (ACM, IEEE)
- Computing is an activity which supports social institutions which in turn are aimed at fulfilling a variety of social functions
- Computing does not appear to be a 'profession' in the strictest sense, the way law and medicine are professions, but it is more of a 'profession' than many other occupations.

# SOFTWARE ENGINEERING

Software engineering appears to be one area of computing that is emerging as a distinct profession within the field of computing.

This trend seems to have been created by the initiative of individuals in the field concerned about the quality and safety of the software being produced and sold.

Developing software engineering as a distinct field involves several of the activities.

- For example, it means identifying a unique body of knowledge that a person must possess to be competent software engineer.
- It means developing educational requirements (curriculum) such that the person who meets the requirements is more likely to produce a quality, safe software than someone without training.

# SOFTWARE ENGINEERING

- It means developing mechanisms for licensing of members.
- This will include identifying or creating the proper organization for issuing licenses and identifying requirements for obtaining a license, such as passing an exam or acquiring a certain number of years of experience.
- Another requirement for professionalization is a code of ethics.
- The state of Texas has boldly taken the first steps in this process and has established software engineering licensing in its state.
- The Texas initiative is a serious attempt at setting standards in the field.

# SOFTWARE DEVELOPMENT AS ENGINEERING

The development of software and information systems in general is now usually regarded as a branch of engineering and the people who practice it are considered to be engineers.

- Traditional engineers design and build a wide variety of objects – dams, bridges, aeroplanes, cars, radio and television transmitters and receivers, computers, plants to make fertilizer or plastics, and so on.

# Characteristic of Engineering

There are two constraints that apply to all such activities and which can be regarded as characteristic of engineering:

- First engineering involves designing and building things that must work properly, that is, must meet a set of predetermined requirements concerning their functionality, their performance, and their reliability;
- the process of designing and building the object must be completed within specified constraints of time and budget.

# THE STATUS OF ENGINEERS

- The legal status of the engineering profession varies a lot from one country to another.  however, the position is that:
- it is illegal to call yourself an engineer in a given state unless you are registered with the State Engineers Registration Board;
- it is illegal for a company to use the word ‘engineering’ in its name unless it employs at least one registered engineer;
- academic programmes including the term engineering in their title must be taught mostly by registered engineers;
- it is illegal to carry out engineering work except under the supervision of a registered engineer.

# Washington Accord

- The Washington Accord is an international agreement that plays a crucial role in the mutual recognition of engineering degrees across various countries.
- The Washington Accord was established in 1989 to promote the mutual recognition of engineering degree programs accredited by its signatory countries.
- The main goal is to ensure that engineering graduates from accredited programs are recognized as meeting the academic requirements for entry into the engineering profession in any of the signatory countries.

# Washington Accord

- The Accord primarily covers undergraduate engineering programs (bachelor's level) that prepare graduates to enter the engineering profession.
- The programs accredited by signatory bodies must meet certain standards that ensure the quality and rigor of engineering education, focusing on outcomes-based education, where the emphasis is on what students are expected to know and be able to do by the time they graduate.